DGMS Examination 2019 — Question 45 Detailed Solution

Junior Scientific Officer (Geology) | Crystallography & Crystal Morphology

QUESTION 45

In each crystal class there is a form, the faces of which intersect all the crystallographic axes at different lengths. This form is known as:

(a) Special form	**(b) General form** CORRECT ANSWER
(c) Open form	(d) Closed form

1. Executive Summary & Answer Verification

The correct option is **(b) General form**.

In morphological crystallography, a **form** (designated by curly braces $\{hkl\}$) is defined as a group of crystal faces that are related to one another by the symmetry operations of the crystal's point group. In every one of the 32 crystal classes (point groups), forms are categorized into two fundamental geometric types:

- **General Form:** A form whose representative face intersects *all* crystallographic axes (a, b, c) at different, non-zero lengths (i.e., intercepts $a/h \neq b/k \neq c/l$ where $h \neq k \neq l \neq 0$). The faces of a general form are *neither parallel nor perpendicular to any symmetry axis or mirror plane*. Because it occupies an arbitrary ("general") angular orientation relative to symmetry elements, applying the point group's symmetry operations generates the **maximum possible number of faces** for that class. The crystal class itself is traditionally named after its general form (e.g., the *Hexoctahedral class* in $4/m\bar{3}2/m$, the *Dipyramidal class* in $4/m$, etc.).
- **Special Form:** A form whose faces occupy a special geometric orientation relative to the symmetry elements (e.g., exactly parallel or perpendicular to a mirror plane or rotation axis). In index notation, at least one index is zero (e.g., $\{100\}$, $\{110\}$) or two or more indices are equal (e.g., $\{111\}$ in cubic). Special forms contain fewer faces than the general form because faces lying on symmetry elements merge or repeat into identical spatial orientations.
- **Open Form vs. Closed Form:** Refers to spatial enclosure. An *open form* (such as a pinacoid, pedion, prism, or dome) does not enclose space by itself and can only exist on a crystal in combination with other forms. A *closed form* (such as an octahedron, dipyramid, rhombohedron, or trapezohedron) encloses space completely and can theoretically form an independent, freestanding crystal. Both open and closed forms can be either general or special depending on their orientation.

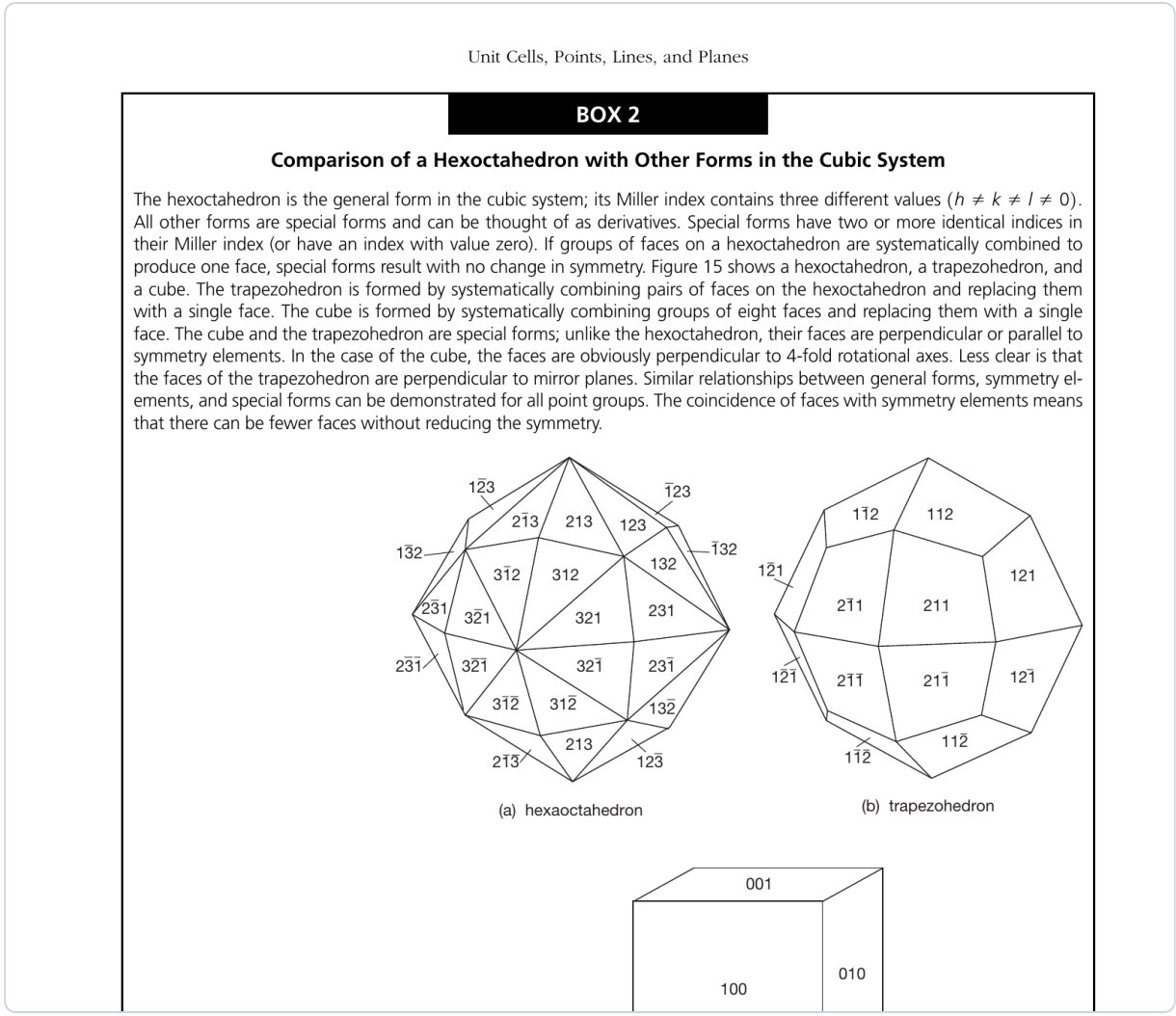


Figure 45.1: *General Form (Hexoctahedron, 48 faces) vs. Special Forms (Trapezohedron {211}, Cube {100}) for cubic point group 4/m 3 2/m. Note that the hexoctahedron faces intersect all three axes at different lengths ($h \neq k \neq l \neq 0$) and do not lie on any symmetry element, yielding the maximum 48 faces. (Dexter Perkins, Mineralogy, 3rd ed., Chapter 11, Figure 15).*

2. Detailed Crystallographic Definitions & Classification

2.1 The General Form ({hkl})

A general form has the general Miller index $\{hkl\}$ (or $\{hkil\}$ in the hexagonal system) where:

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h ≠ k ≠ l ≠ 0   and   |h| ≠ |k| ≠ |l|
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Key characteristics of the General Form:

1. **Intersects All Axes:** Every face intersects the a, b, and c crystallographic axes at non-zero, unequal lengths. No face is parallel to any crystallographic axis (no zero index).
2. **No Symmetry Coincidence:** Faces are completely independent of symmetry planes and rotation axes—they are neither perpendicular nor parallel to any mirror plane or rotation axis.
3. **Maximum Multiplicity:** It possesses the highest multiplicity (number of equivalent faces) possible within that point group. For example, in the holohedral cubic class ($4/m\bar{3}2/m$), the general form $\{hkl\}$ produces 48 faces (the *hexoctahedron*).

4. **Class Nomenclature:** Every one of the 32 crystal classes is named after its general form. For instance:
    - Point group $2/m$: **Prismatic class** (general form is a prism $\{hkl\}$).
    - Point group mmm: **Rhombic dipyramidal class** (general form is a rhombic dipyramid $\{hkl\}$).
    - Point group $4/mmm$: **Ditetragonal dipyramidal class** (general form $\{hkl\}$).
    - Point group $6/mmm$: **Dihexagonal dipyramidal class** (general form $\{hkil\}$).
    - Point group $m\bar{3}m$: **Hexoctahedral class** (general form $\{hkl\}$).

2.2 Special Forms

When a face is positioned such that it is perpendicular to a rotation axis, parallel to a rotation axis, or perpendicular to a mirror plane, symmetry operations map faces onto each other, resulting in fewer total faces:

- **Cube $\{100\}$:** In $4/m\bar{3}2/m$, faces are perpendicular to the 4-fold axes. Multiplicity collapses from 48 down to 6 faces.
- **Octahedron $\{111\}$:** Faces intersect all three axes at *equal* lengths ($h = k = l = 1$). Faces are perpendicular to 3-fold rotoinversion axes. Multiplicity is 8 faces.
- **Rhombic Dodecahedron $\{110\}$:** Faces intersect two axes at equal lengths and are parallel to the third axis ($l = 0$). Multiplicity is 12 faces.
- **Trapezohedron $\{hll\}$ or $\{hhl\}$:** (e.g., $\{211\}$). Faces are perpendicular to mirror planes. Multiplicity is 24 faces.

3. Comprehensive Comparison Table of Form Classifications

Form Category	Miller Index Pattern	Axis Intersect Property	Relation to Symmetry Elements	Multiplicity / Examples
General Form	$\{hkl\}$ with $h \neq k \neq l \neq 0$	Intersects all axes at unequal, non-zero lengths	Not parallel or perpendicular to any symmetry axis or plane	Maximum multiplicity of the class (e.g., Hexoctahedron = 48 faces)
Special Form	Indices contain 0, or two/more equal indices ($h=k$, etc.)	Parallel to an axis, or intersects 2+ axes at equal lengths	Parallel or perpendicular to rotation axes or mirror planes	Reduced multiplicity (e.g., Cube {100} = 6 faces; Octahedron {111} = 8 faces)
Open Form	Variable ($\{hkl\}$, $\{hk0\}$, $\{001\}$, etc.)	Does not enclose 3D space by itself	Can be general or special depending on class	Pedion (1), Pinacoid (2), Dome (2), Sphenoid (2), Prism (3, 4, 6, 8, 12), Pyramid
Closed Form	Variable ($\{hkl\}$, $\{hkl\}$ in high-symmetry classes)	Encloses 3D space completely	Can be general or special depending on class	Dipyramid, Rhombohedron, Scalenohedron, Trapezohedron, Octahedron, Cube

4. Evaluation of All Multiple Choice Options

- **(a) Special form — INCORRECT:** Special forms have faces that intersect axes at special values (e.g., equal lengths or parallel to axes with index 0), coinciding with symmetry planes or axes.

- **(b) General form — CORRECT:** Exactly matches the definition. In every crystal class, the general form has faces intersecting all crystallographic axes at different lengths ($h \neq k \neq l \neq 0$), yielding the highest face count and defining the class name.
- **(c) Open form — INCORRECT:** An open form is defined by whether its faces completely enclose space, not by how its faces intersect the axes. For example, pinacoids and prisms are open forms, but general forms in many low-symmetry classes (e.g., triclinic, monoclinic) are open forms, while in higher symmetry classes they are closed forms.
- **(d) Closed form — INCORRECT:** A closed form encloses space completely. Many special forms (cube, octahedron, dodecahedron) are closed forms, but they do NOT intersect all axes at different lengths.

Standard Textbook Reference

Dexter Perkins, *Mineralogy* (3rd Edition, Chapter 11: Unit Cells, Points, Lines, and Planes, pp. 320–324):
"The hexoctahedron is the general form in the cubic system; its Miller index contains three different values {hkl}. All other forms are special forms and can be thought of as derivatives... Faces of a general form are neither parallel nor perpendicular to any symmetry element, while faces of special forms are... In each crystal class, the general form is the one with {hkl} that intersects all crystallographic axes at different lengths."